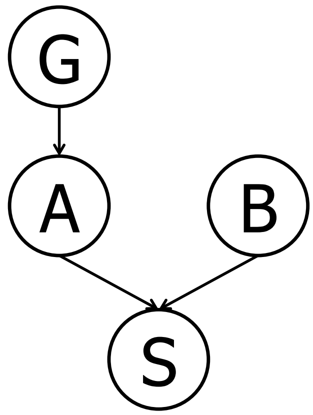
Q3. [14 pts] Bayes' Nets Representation and Probability

Suppose that a patient can have a symptom (S) that can be caused by two different diseases (A and B). It is known that the variation of gene G plays a big role in the manifestation of disease A. The Bayes' Net and corresponding conditional probability tables for this situation are shown below. For each part, you may leave your answer as an arithmetic expression.

$\mathbb{P}(G)$	
$+g$	0.1
$-g$	0.9

$\mathbb{P}(A\|G)$		
$+g$	$+a$	1.0
$+g$	$-a$	0.0
$-g$	$+a$	0.1
$-g$	$-a$	0.9



$\mathbb{P}(B)$	
$+b$	0.4
$-b$	0.6

$\mathbb{P}(S\|A, B)$			
$+a$	$+b$	$+s$	1.0
$+a$	$+b$	$-s$	0.0
$+a$	$-b$	$+s$	0.9
$+a$	$-b$	$-s$	0.1
$-a$	$+b$	$+s$	0.8
$-a$	$+b$	$-s$	0.2
$-a$	$-b$	$+s$	0.1
$-a$	$-b$	$-s$	0.9

(a) [2 pts] Compute the following entry from the joint distribution:

$\mathbb{P}(+g, +a, +b, +s) =$

$\mathbb{P}(+g)\mathbb{P}(+a|+g)\mathbb{P}(+b)\mathbb{P}(+s|+b, +a) = (0.1)(1.0)(0.4)(1.0) = 0.04$

(b) [2 pts] What is the probability that a patient has disease A?

$\mathbb{P}(+a) =$ $\mathbb{P}(+a|+g)\mathbb{P}(+g) + \mathbb{P}(+a|-g)\mathbb{P}(-g) = (1.0)(0.1) + (0.1)(0.9) = 0.19$

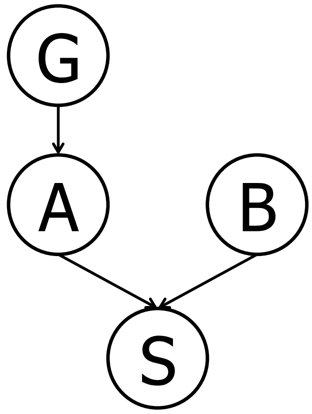
(c) [2 pts] What is the probability that a patient has disease A given that they have disease B?

$\mathbb{P}(+a|+b) =$ $\mathbb{P}(+a) = 0.19$ The first equality holds true as we have $A \perp\!\!\!\perp B$, which can be inferred from the graph of the Bayes' net.

The figures and table below are identical to the ones on the previous page and are repeated here for your convenience.

$\mathbb{P}(G)$	
$+g$	0.1
$-g$	0.9

$\mathbb{P}(A\|G)$		
$+g$	$+a$	1.0
$+g$	$-a$	0.0
$-g$	$+a$	0.1
$-g$	$-a$	0.9



$\mathbb{P}(B)$	
$+b$	0.4
$-b$	0.6

$\mathbb{P}(S\|A,B)$			
$+a$	$+b$	$+s$	1.0
$+a$	$+b$	$-s$	0.0
$+a$	$-b$	$+s$	0.9
$+a$	$-b$	$-s$	0.1
$-a$	$+b$	$+s$	0.8
$-a$	$+b$	$-s$	0.2
$-a$	$-b$	$+s$	0.1
$-a$	$-b$	$-s$	0.9

(d) [4 pts] What is the probability that a patient has disease A given that they have symptom S and disease B?

$$\mathbb{P}(+a|+s,+b) = \frac{\mathbb{P}(+a,+b,+s)}{\mathbb{P}(+a,+b,+s)+\mathbb{P}(-a,+b,+s)} = \frac{\mathbb{P}(+a)\mathbb{P}(+b)\mathbb{P}(+s|+a,+b)}{\mathbb{P}(+a)\mathbb{P}(+b)\mathbb{P}(+s|+a,+b)+\mathbb{P}(-a)\mathbb{P}(+b)\mathbb{P}(+s|-a,+b)}$$
$$= \frac{(0.19)(0.4)(1.0)}{(0.19)(0.4)(1.0)+(0.81)(0.4)(0.8)} = \frac{0.076}{0.076+0.2592} \approx 0.2267$$

(e) [2 pts] What is the probability that a patient has the disease carrying gene variation G given that they have disease A?

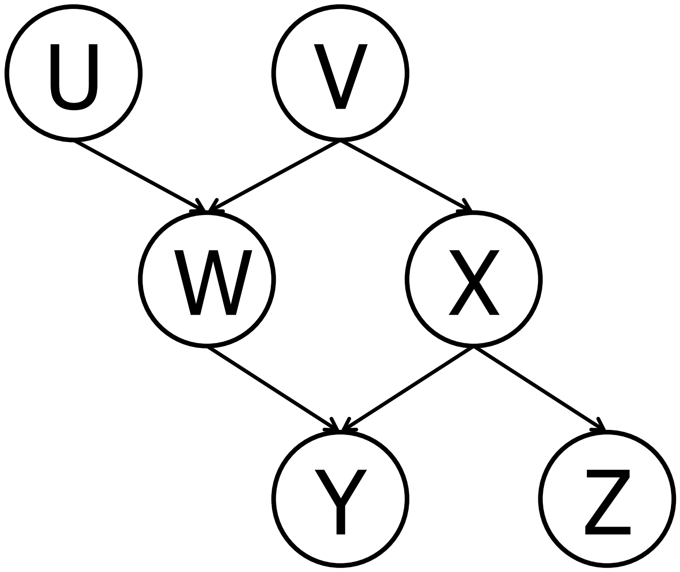
$$\mathbb{P}(+g|+a) = \frac{\mathbb{P}(+g)\mathbb{P}(+a|+g)}{\mathbb{P}(+g)\mathbb{P}(+a|+g)+\mathbb{P}(-g)\mathbb{P}(+a|-g)} = \frac{(0.1)(1.0)}{(0.1)(1.0)+(0.9)(0.1)} = \frac{0.1}{0.1+0.09} = 0.5263$$

(f) [2 pts] What is the probability that a patient has the disease carrying gene variation G given that they have disease B?

$\mathbb{P}(+g|+b) = \mathbb{P}(+g) = 0.1$ The first equality holds true as we have $G \perp\!\!\!\perp B$, which can be inferred from the graph of the Bayes' net.

Q4. [16 pts] D-Separation

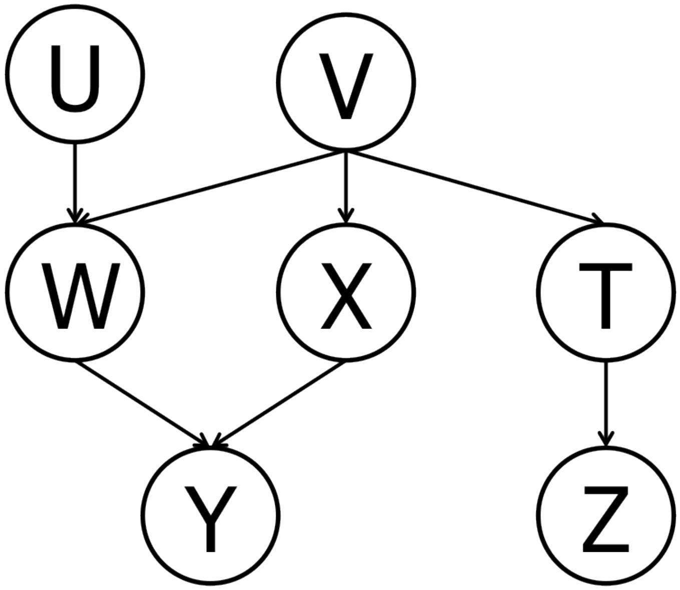
(a) [16 pts] Based only on the structure of the (new) Bayes' Net given below, circle whether the following conditional independence assertions are guaranteed to be true, guaranteed to be false, or cannot be determined by the structure alone.



$U \perp\!\!\!\perp V$	$\boxed{\text{Guaranteed true}}$	Cannot be determined	Guaranteed false
$U \perp\!\!\!\perp V \mid W$	Guaranteed true	$\boxed{\text{Cannot be determined}}$	Guaranteed false
$U \perp\!\!\!\perp V \mid Y$	Guaranteed true	$\boxed{\text{Cannot be determined}}$	Guaranteed false
$U \perp\!\!\!\perp Z \mid W$	Guaranteed true	$\boxed{\text{Cannot be determined}}$	Guaranteed false
$U \perp\!\!\!\perp Z \mid V, Y$	Guaranteed true	$\boxed{\text{Cannot be determined}}$	Guaranteed false
$U \perp\!\!\!\perp Z \mid X, W$	$\boxed{\text{Guaranteed true}}$	Cannot be determined	Guaranteed false
$W \perp\!\!\!\perp X \mid Z$	Guaranteed true	$\boxed{\text{Cannot be determined}}$	Guaranteed false
$V \perp\!\!\!\perp Z \mid X$	$\boxed{\text{Guaranteed true}}$	Cannot be determined	Guaranteed false

Q5. [22 pts] Variable Elimination

(a) [10 pts] For the Bayes' net below, we are given the query $P(Z \mid +y)$. All variables have binary domains. Assume we run variable elimination to compute the answer to this query, with the following variable elimination ordering: U, V, W, T, X.



Complete the following description of the factors generated in this process:

After inserting evidence, we have the following factors to start out with:

$$P(U), P(V), P(W|U,V), P(X|V), P(T|V), P(+y|W,X), P(Z|T).$$

When eliminating U we generate a new factor f_1 as follows:

$$f_1(V,W) = \sum_u P(u)P(W|u,V).$$

This leaves us with the factors:

$$P(V), P(X|V), P(T|V), P(+y|W,X), P(Z|T), f_1(V,W).$$

When eliminating V we generate a new factor f_2 as follows:

$$f_2(T,W,X) = \sum_v P(v)P(X|v)P(T|v)f_1(v,W).$$

This leaves us with the factors:

$$P(+y|W,X), P(Z|T), f_2(T,W,X).$$

When eliminating W we generate a new factor f_3 as follows:

$$f_3(T, X, +y) = \sum_w P(+y|w, X) f_2(T, w, X).$$

This leaves us with the factors:

$$P(Z|T), f_3(T, X, +y).$$

When eliminating T we generate a new factor f_4 as follows:

$$f_4(X, +y, Z) = \sum_t P(Z|t) f_3(t, X, +y).$$

This leaves us with the factor:

$$f_4(X, +y, Z).$$

When eliminating X we generate a new factor f_5 as follows:

$$f_5(+y, Z) = \sum_x f_4(x, +y, Z).$$

This leaves us with the factor:

$$f_5(+y, Z)$$

.

(b) [2 pts] Briefly explain how $P(Z \mid +y)$ can be computed from f_5.

Simply renormalize f_5 to obtain $P(Z \mid +y)$. Concretely, $P(z \mid +y) = \frac{f_5(z,+y)}{\sum_{z'} f_5(z',y)}$.

(c) [2 pts] Amongst $f_1, f_2, \ldots, f_5$, which is the largest factor generated? (Assume all variables have binary domains.) How large is this factor?

$f_2(T, W, X)$ is the largest factor generated. It has 3 variables, hence $2^3 = 8$ entries.

(d) [8 pts] Find a variable elimination ordering for the same query, i.e., for $P(Z \mid y)$, for which the maximum size factor generated along the way is smallest. Hint: the maximum size factor generated in your solution should have only 2 variables, for a size of $2^2 = 4$ table. Fill in the variable elimination ordering and the factors generated into the table below.

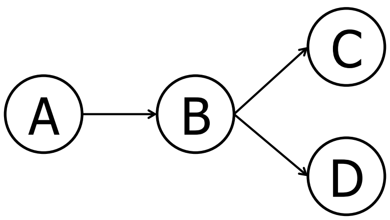
Note: in the naive ordering we used earlier, the first line in this table would have had the following two entries: U, $f_1(V, W)$.

Note: multiple orderings are possible.

Variable Eliminated	Factor Generated
T	$f_1(Z, V)$
X	$f_2(+y, W, V)$
W	$f_3(+y, U, V)$
U	$f_4(+y, V)$
V	$f_5(+y, Z)$

Q6. [7 pts] Bayes' Nets Sampling

Assume the following Bayes net, and the corresponding distributions over the variables in the Bayes net:



A	$\mathbb{P}(A)$
$+a$	$1/5$
$-a$	$4/5$

A	B	$\mathbb{P}(B\|A)$
$+a$	$+b$	$1/5$
$+a$	$-b$	$4/5$
$-a$	$+b$	$1/2$
$-a$	$-b$	$1/2$

B	C	$\mathbb{P}(C\|B)$
$+b$	$+c$	$1/4$
$+b$	$-c$	$3/4$
$-b$	$+c$	$2/5$
$-b$	$-c$	$3/5$

B	D	$\mathbb{P}(D\|B)$
$+b$	$+d$	$1/2$
$+b$	$-d$	$1/2$
$-b$	$+d$	$4/5$
$-b$	$-d$	$1/5$

(a) [2 pts] Your task is now to estimate $\mathbb{P}(+b| - a, -c, -d)$ using rejection sampling. Below are some samples that have been produced by prior sampling (that is, the rejection stage in rejection sampling hasn't happened yet). Cross out the samples that would be rejected by rejection sampling:

$$\cancel{-a \quad -b \quad +c \quad +d} \qquad\qquad -a \quad -b \quad -c \quad -d$$
$$\cancel{+a \quad -b \quad -c \quad +d} \qquad\qquad \cancel{-a \quad +b \quad +c \quad +d}$$
$$\cancel{-a \quad -b \quad +c \quad -d} \qquad\qquad \cancel{+a \quad -b \quad -c \quad -d}$$

(b) [1 pt] Using those samples, what value would you estimate for $\mathbb{P}(+b| - a, -c, -d)$ using rejection sampling?

0

(c) [4 pts] Using the following samples (which were generated using likelihood weighting), estimate $\mathbb{P}(+b \mid -a, -c, -d)$ using likelihood weighting, or state why it cannot be computed.

$$-a \quad -b \quad -c \quad -d$$
$$-a \quad +b \quad -c \quad -d$$
$$-a \quad -b \quad -c \quad -d$$

We compute the weights of each solution, which are the product of the probabilities of the evidence variables conditioned on their parents.

$$w_1 = w_3 = P(-a)P(-c \mid -b)P(-d \mid -b) = 4/5 * 3/5 * 1/5 = 12/125$$
$$w_2 = P(-a)P(-c \mid +b)P(-d \mid +b) = 4/5 * 3/4 * 1/2 = 12/40$$

so normalizing, we have $(w_2)/(w_2 + w_1 + w_3) = \frac{12/40}{12/40+12/125+12/125} = \frac{25}{41} \approx 0.609756$.

7. (12 points) Mumps Outbreak There has been an outbreak of mumps at UC Berkeley. You feel fine, but you're worried that you might already be infected and therefore won't be healthy enough to take your CS 188 midterm. You decide to use Bayes nets to analyze the probability that you've contracted the mumps.

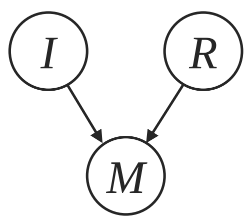
You first think about the following two factors:

- You think you have immunity from the mumps ($+i$) due to being vaccinated recently, but the vaccine is not completely effective, so you might not be immune ($-i$).
- Your roommate didn't feel well yesterday, and though you aren't sure yet, you suspect they might have the mumps ($+r$).

Denote these random variables by I and R. Let the random variable M take the value $+m$ if you have the mumps, and $-m$ if you do not. You write down the following Bayes net to describe your chances of being sick:

I	P(I)
+i	0.8
-i	0.2

R	P(R)
+r	0.4
-r	0.6



I	R	M	P(M|I,R)
+i	+r	+m	0
+i	+r	-m	1.0
+i	-r	+m	0
+i	-r	-m	1.0
-i	+r	+m	0.7
-i	+r	-m	0.3
-i	-r	+m	0.2
-i	-r	-m	0.8

(a) (2 pt) Fill in the following table with the joint distribution over I, M, and R, $P(I, M, R)$.

I	R	M	$P(I, R, M)$
$+i$	$+r$	$+m$	0
$+i$	$+r$	$-m$	0.32
$+i$	$-r$	$+m$	0
$+i$	$-r$	$-m$	0.48
$-i$	$+r$	$+m$	0.056
$-i$	$+r$	$-m$	0.024
$-i$	$-r$	$+m$	0.024
$-i$	$-r$	$-m$	0.096

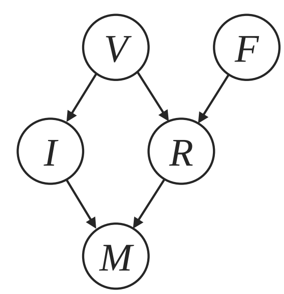
(b) (2 pt) What is the marginal probability $P(+m)$ that you have the mumps?

$$P(+m) = \sum_{i,r} P(i, r, +m) = P(+i, +r, +m) + P(+i, -r, +m) + P(-i, +r, +m) + P(-i, -r, +m)$$
$$= 0 + 0 + 0.056 + 0.024 = 0.08 = \frac{8}{100}$$

(c) (3 pt) Assuming you do have the mumps, you're concerned that your roommate may have the disease as well. What is the probability $P(+r \mid +m)$ that your roommate has the mumps given that you have the mumps? Note that you still don't know whether or not you have immunity.

$$P(+r \mid +m) = \frac{P(+r, +m)}{P(+m)} = \frac{\sum_i P(i, +r, +m)}{P(+m)} = \frac{0 + 0.056}{0.080} = 0.143 = \frac{7}{10}$$

You're still not sure if you have enough information about your chances of having the mumps, so you decide to include two new variables in the Bayes net. Your roommate went to a frat party over the weekend, and there's some chance another person at the party had the mumps ($+f$). Furthermore, both you and your roommate were vaccinated at a clinic that reported a vaccine mix-up. Whether or not you got the right vaccine ($+v$ or $-v$) has ramifications for both your immunity (I) and the probability that your roommate has since contracted the disease (R). Accounting for these, you draw the modified Bayes net shown on the right.



(d) (5 pt) Circle all of the following statements which are *guaranteed* to be true for this Bayes net:

(i) $V \perp\!\!\!\perp M \mid I, R$

(ii) $V \perp\!\!\!\perp M \mid R$

(iii) $M \perp\!\!\!\perp F \mid R$

(iv) $V \perp\!\!\!\perp F$

(v) $V \perp\!\!\!\perp F \mid M$

(vi) $V \perp\!\!\!\perp F \mid I$

(i), (iv), (vi). Use the Bayes ball algorithm.